

THE ZCASH ARBORETUM

Zcash for Bitcoiners

the mathematics of a private transaction

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Arithmetic with a modulus

For a prime p , the field $\mathbb{F}_p$ is arithmetic modulo p .

$$\mathbb{F}_{97} = \{0, 1, \dots, 96\}.$$

$$96 + 2 = 1, \quad 5 \cdot 39 = 1.$$

Every nonzero element has an inverse:

$$5^{-1} = 39 \quad \text{in } \mathbb{F}_{97}.$$

Curve points and scalars are different objects

The Pallas curve has equation $y^2 = x^3 + 5$ over $\mathbb{F}_p$. Its points, including the identity O , form a group.

$$[k]P = \underbrace{P + \cdots + P}_{k \text{ additions}}, \quad [a + b]P = [a]P + [b]P.$$

$$\underbrace{p = p_{\text{Pallas}}}_{\text{coordinates}}$$

$$\underbrace{q = p_{\text{Vesta}}}_{\text{group order; scalars in } \mathbb{F}_q}$$

Commit to an amount

Fix public, nonzero curve points V, R . Assume nobody knows λ (lambda) with $R = [\lambda]V$.

$$\boxed{C(v;r) = [v]V + [r]R}$$

The amount is v ; the fresh, secret blinder is $r \in \mathbb{F}_q$.

hiding
cannot read v

binding
cannot change the opening (v,r)

Why it hides

Multiplication by R permutes all q group points:

$$r \text{ uniform in } \mathbb{F}_q \implies [r]R \text{ uniform in the group.}$$

Adding a fixed point only permutes them again:

$$[v]V + [r]R \text{ has the same distribution for every } v.$$

Why it binds

Suppose two different amounts open the same commitment:

$$[v_0]V + [r_0]R = [v_1]V + [r_1]R.$$

Then

$$[v_0 - v_1]V = [r_1 - r_0]R$$

and therefore

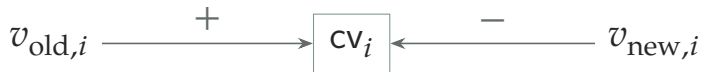
$$\boxed{\lambda = (v_0 - v_1)(r_1 - r_0)^{-1} \pmod{q}.}$$

That would reveal the supposedly unknown relation $R = [\lambda]V$.

One Action: one input, one output

An *Action* pairs an old note with a new note. A note is a private coin; either side may carry zero value.

$$cv_i = [v_{\text{old},i} - v_{\text{new},i}]V + [r_i]R.$$



public commitment; hidden amounts and blinder

Conservation is a cancellation

Let b be the public net value leaving the shielded pool. The public binding key is

$$\text{bvk} = \sum_i \text{cv}_i - [b]V.$$

Expanding:

$$\text{bvk} = \underbrace{\left[\sum_i (v_{\text{old},i} - v_{\text{new},i}) - b \right]}_{\Delta \text{ (delta): imbalance}} V + \left[\sum_i r_i \right] R.$$

Why the binding signature enforces balance

A binding signature requires knowing k with $\text{bvk} = [k]R$. Let $s := \sum_i r_i$ be the total blinder. Then

$$[\Delta]V = [k - s]R.$$

If balanced, $\Delta = 0$ and the author knows $k = s$.

$$\Delta \neq 0 \implies \lambda = \Delta(k - s)^{-1},$$

revealing the forbidden relation $R = [\lambda]V$.

Field equality is not yet integer equality

In a field,

$$q - 1 = -1, \quad q = 0.$$

The proof therefore constrains each note amount to

$$0 \leq v < 2^{64}.$$

Bounded values and transaction size prevent a nonzero integer imbalance from wrapping around modulo q .

A recipient is a curve relation

A diversifier d selects a public base g_d by hashing to the curve. The private incoming-viewing scalar is ivk .



$$\text{addr} = (d, pk_d).$$

The private note

$$\mathbf{n} = (\text{addr}, v, \rho, \psi, \text{rcm}).$$

$\text{addr} = (d, \text{pk}_d)$ recipient

v amount in zatoshi

ρ (rho) uniqueness seed, in $\mathbb{F}_p$

ψ (psi) additional note randomness, in $\mathbb{F}_p$

rcm commitment blinder, in $\mathbb{F}_q$

One ZEC is 10^8 zatoshi.

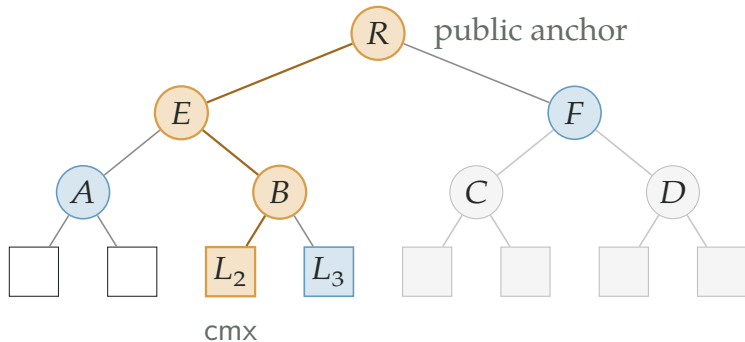
From a note to a public leaf

Write P^* for an encoded point; $\parallel$ joins bit strings. The encoding LE_k has k bits, lowest bit first.

$$\begin{aligned} m &= g_d^* \parallel \text{pk}_d^* \parallel \text{LE}_{64}(v) \\ &\quad \parallel \text{LE}_{255}(\rho) \parallel \text{LE}_{255}(\psi), \\ \text{cm} &= \text{NoteCommit}_{\text{rcm}}(m), \\ \text{cmx} &= \text{Extract}(\text{cm}). \end{aligned}$$

The note commitment cm is a point; cmx is its x -coordinate. Unlike cv_i , NoteCommit binds the note's full contents.

Membership without revealing the leaf



Public tree; hidden choice of leaf and path.
Depth-3 illustration; actual depth 32.

Three hashes reconstruct the root

In the illustrated tree, H_i hashes two children at height i .

$$B = H_0(\underbrace{L_2, L_3}_{\text{cmx}}),$$

$$E = H_1(A, B),$$

$$R = H_2(E, F).$$

The hidden position is 010_2 ; the siblings are (L_3, A, F) .

$R = \text{anchor.}$

The spent-note marker is fixed

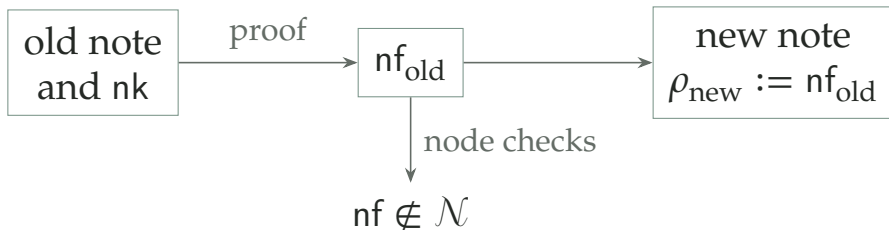
The owner's secret nullifier key is nk . Define $F_{nk}(\rho) = \text{PoseidonHash}(nk, \rho)$, a keyed field hash; G^{nf} is a fixed public curve point.

$$nf = \text{Extract}([F_{nk}(\rho) + \psi]G^{nf} + cm).$$

The sum is taken in $\mathbb{F}_p$, then read as a scalar; $p < q$.

Same note and key $\implies$ same *nullifier* nf .

The proof and the node check different things



The set $\mathcal{N}$ contains spent nullifiers. Reject repeats; insert new ones only when accepting the transaction.

The hidden owner links both sides

Let ask be the signing secret and $G^{\text{SpendAuth}}$ its fixed public base.
The validating key is

$$ak = [ask]G^{\text{SpendAuth}}.$$

A key commitment with blinder r_{ivk} links it to the address:

$$ivk = \text{Commitlvk}_{r_{ivk}}(\text{Extract}(ak), nk), \quad pk_d = [ivk]g_d.$$

Same hidden ak, nk ; same note recipient.

Randomise the public signing key

Choose a fresh secret α (alpha), in $\mathbb{F}_q$.

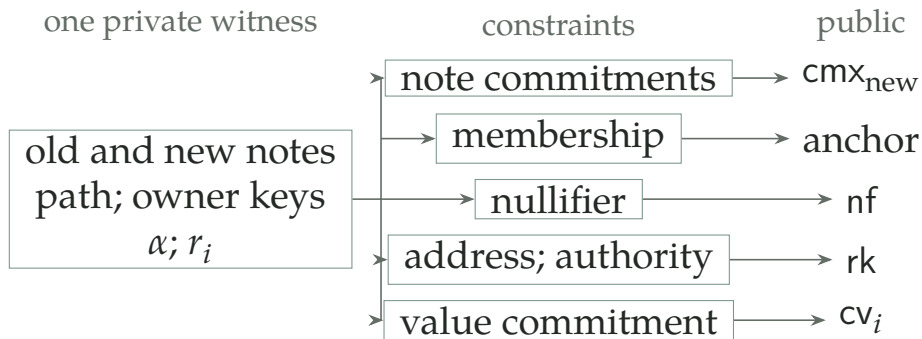
$$\text{rsk} = \text{ask} + \alpha,$$

$$\begin{aligned}\text{rk} &= \text{ak} + [\alpha]G^{\text{SpendAuth}} \\ &= [\text{rsk}]G^{\text{SpendAuth}}.\end{aligned}$$

The proof links rk to the hidden owner.

The spend signature authorises the transaction under rk.

The mathematical statement, not the secret



A *witness* is the secret data satisfying the public claim. Every check uses the same notes and owner keys.

A small claim with the same proof machinery

Use $\mathbb{F}_{97}$ as a toy circuit field. The public claim is:

there exists x such that $x^3 + x + 5 = 35$.

Our private witness is $x = 3$.

$$3 \xrightarrow{\text{square}} 9 \xrightarrow{\times 3} 27 \xrightarrow{+3} 30 \xrightarrow{+5} 35.$$

Turn the computation into rows

The *trace* records intermediate values in columns a, b, c .

row	a	b	c	constraint
0	3	3	9	$ab - c = 0$
1	9	3	27	$ab - c = 0$
2	27	3	30	$a + b - c = 0$
3	30	0	0	$a + 5 - 35 = 0$

These private cells are called *advice*.

One gate, selected differently on each row

A *gate* is an equation imposed on a row:

$$q_M ab + q_L a + q_R b + q_O c + q_C + \text{PI} = 0.$$

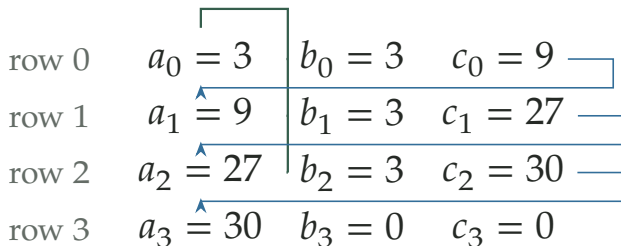
The q 's are fixed public coefficients, called *selectors*. The column PI carries the public input.

row	q_M	q_L	q_R	q_O	q_C	PI
0,1	1	0	0	-1	0	0
2	0	1	1	-1	0	0
3	0	1	0	0	5	-35

Equal values must actually be wired together

Write a_i for column a , row i , and likewise for b, c .

row 0	$a_0 = 3$	$b_0 = 3$	$c_0 = 9$
row 1	$a_1 = 9$	$b_1 = 3$	$c_1 = 27$
row 2	$a_2 = 27$	$b_2 = 3$	$c_2 = 30$
row 3	$a_3 = 30$	$b_3 = 0$	$c_3 = 0$

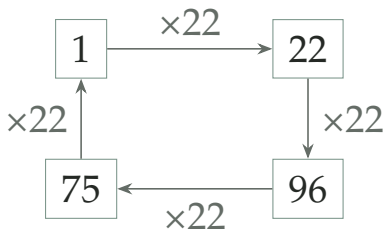


These *copy constraints* are separate from the row gates.

Give the rows multiplicative labels

Choose $\omega = 22$ (omega), with $\omega^2 = -1$ and $\omega^4 = 1$.

$$H = \{1, 22, 96, 75\} \subset \mathbb{F}_{97}.$$



row i gets label ω^i
a cycle of field values,
not a plot in the plane

A column becomes a polynomial

A *polynomial* is a sum of coefficients times powers of X . Its *degree* is the highest nonzero power.

$$a(X) = 90 + 61X + 22X^2 + 24X^3.$$

label h	1	22	96	75
$a(h)$	3	9	27	30

Interpolation: four values at distinct labels determine a unique polynomial of degree < 4 .

Turn all row gates into one polynomial

Interpolate every column, including the selectors and PI.
Define the gate residual:

$$\begin{aligned} G(X) = & q_M(X)a(X)b(X) \\ & + q_L(X)a(X) + q_R(X)b(X) \\ & + q_O(X)c(X) + q_C(X) + \text{PI}(X). \end{aligned}$$

All row gates hold exactly when

$$G(h) = 0 \quad \text{for every } h \in H.$$

Many checks become one divisibility claim

The *vanishing polynomial* has the row labels as roots (zeros):

$$Z_H(X) := \prod_{h \in H} (X - h) = X^4 - 1.$$

Hence

$$G(h) = 0 \text{ for all } h \in H \iff G(X) = t(X)Z_H(X)$$

for a polynomial quotient t .

$$\deg G \leq 9, \quad \deg t \leq 5.$$

Change one cell; the identity breaks

Replace only $c_0 = 9$ by $c_0 = 10$. The first row now has residual

$$3 \cdot 3 - 10 = -1 = 96 \quad \text{in } \mathbb{F}_{97}.$$

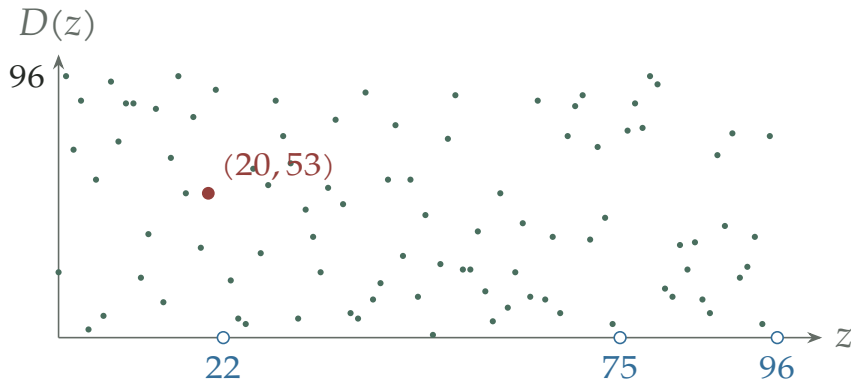
Dividing the altered G^* by Z_H leaves

$$G^* = t^*Z_H + D,$$

$$D(X) = 24(1 + X + X^2 + X^3) \neq 0.$$

A nonzero polynomial has few roots

For this remainder, the only zeros are 22, 75, 96.



For uniform $z \in \mathbb{F}_{97}$, this cheat passes with probability $3/97$.

One unpredictable point tests the identity

Fix the polynomials first; then sample z . At $z = 20$, the verifier checks

$$G(z) \stackrel{?}{=} t(z)(z^4 - 1).$$

$$\text{honest: } 75 = 67 \cdot 46 = 75 \pmod{97},$$

$$\text{altered: } 20 \neq 64 \pmod{97}.$$

For any nonzero residual of degree at most 9, the acceptance probability is at most $9/97$.

Commit to the polynomial before the challenge

In the real scheme: coefficients in $\mathbb{F}_p$; points on Vesta, whose group order is p .

$$f(X) = \sum_{i=0}^{d-1} f_i X^i,$$

$$C_f = \sum_{i=0}^{d-1} [f_i]G_i + [r]B.$$

The public points G_i, B have no known scalar relation; the fresh blinder is $r \in \mathbb{F}_p$. An *evaluation proof* establishes $f(z) = y$ for the polynomial fixed by C_f .

Evaluation is an inner product

An *inner product* multiplies corresponding entries and adds:

$$\langle (a_0, \dots, a_{d-1}), (b_0, \dots, b_{d-1}) \rangle := \sum_i a_i b_i.$$

So

$$f(z) = \langle (f_0, \dots, f_{d-1}), (1, z, \dots, z^{d-1}) \rangle.$$

Halo 2 proves this committed inner-product claim by repeatedly halving the vectors.

The toy verifier's check

Prove the evaluations against their commitments:

$$a(z), \quad b(z), \quad c(z), \quad t(z).$$

The verifier computes $G(z)$ from these values and the public selectors and $PI(z)$, then checks

$$G(z) = t(z)(z^4 - 1).$$

The prover cannot substitute an unrelated G .

The order is part of the proof



The *transcript* is the public statement and messages so far. The *Fiat–Shamir transform* derives each challenge by hashing it:

$$z = \text{HashToField}(\text{transcript}).$$

Anyone can replay the checks without an interactive verifier.

Binding does not by itself give privacy

Unblinded evaluations are equations in secret coefficients.

$$f(z) = f_0 + f_1z + \cdots + f_{d-1}z^{d-1}.$$

Halo 2 adds randomness in two places:

random unused rows
mask evaluations

random commitment blinders
hide coefficients

Zero knowledge: the proof reveals no extra witness information beyond the public statement.

Back to the transaction

Proof	hidden notes satisfy the equations
Spend signatures	owners authorise this transaction
Binding signature	committed amounts balance
Node state	anchor allowed; nullifiers unused

No single equation replaces the other checks.